Place value

1 Write down the place value represented by the digit 7 in the following numbers:

a 48.07 **b** 0.487 **c** 47.008 **d** 40.708

2 How do you say the following numbers?

a 59.412 **b** 6.026

3 Write the number 'sixty five and six hundred and eighty nine thousandths' as a decimal.

4 Write the following numbers in decimal form:

a Four tenths
b Seventy hundredths
c Seventy six hundredths
d Two and three tenths
e Six and seven hundredths
f Twelve and twenty-two thousandths

5 Write the following fractions as decimals:

a $\frac{7}{10}$ **b** $\frac{94}{100}$ **c** $\frac{61}{1000}$ **d** $\frac{689}{1000}$
e $\frac{2}{10}$ **f** $\frac{9}{100}$ **g** $\frac{7}{1000}$ **h** $\frac{63}{1000}$

6 For each of the following, fill in the boxes with the missing numbers:

a $8.43 = 8 + \square + 0.03$ **b** $29.65 = 20 + 9 + \square + 0.05$
c $8.52 = 8 + \square$ **d** $31.305 = 30 + 1 + \square + \square$

Expanded form

7 Convert the following numbers from expanded form to decimal form:

a $2 \times 10 + 5 \times 1 + 7 \times \frac{1}{10} + 3 \times \frac{1}{100} + 2 \times \frac{1}{1000}$
b $1 \times 10 + 0 \times 1 + 2 \times \frac{1}{10} + 0 \times \frac{1}{100} + 1 \times \frac{1}{1000}$
c $3 \times 100 + 0 \times 10 + 3 \times 1 + 0 \times \frac{1}{10} + 1 \times \frac{1}{100} + 4 \times \frac{1}{1000}$
d $4 \times 1 + 0 \times \frac{1}{10} + 0 \times \frac{1}{100} + 1 \times \frac{1}{1000}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Consider the number read as " 34 and 695 thousandths":

a Write the number in expanded form.

b Write the number in decimal form.

9 Write the following numbers in decimal form:

a $7 + \frac{4}{10} + \frac{2}{100}$

b $75 + \frac{8}{10} + \frac{2}{100}$

c $500 + 2 + \frac{8}{10} + \frac{3}{1000}$

d $12 + \frac{2}{100} + \frac{3}{10000}$

Compare decimals

10 For each of the following, choose the larger decimal number:

a 0.71 or 0.7

b 0.39 or 0.4

c 2.305 or 2.503

d 4.9999 or 4.8

11 Which is the smaller decimal: 0.76 or 0.67?

12 Write the following numbers in ascending order:

a $0.24, 0.204, 2.4, 0.024, 0.0024$

b $3.68, 3.86, 3.068, 3.681, 3.1$

13 Write the following numbers in descending order:

a $0.54, 0.504, 0.5, 0.054, 0.0054$

b $7.28, 7.82, 7.078, 7.168, 7.8$

14 For each of the following, insert the symbol $>$ or $<$ to make the expression true:

a $0.71 \square 0.7$

b $0.39 \square 0.389$

c $2.305 \square 2.309$

d $4.9998 \square 4.9999$

e $0.001 \square 0.0009$

f $2.189 \square 2.19$

g $0.035 \square 0.0043$

h $0.999\,99 \square 1$

Multiply and divide by powers of 10

15 Consider the product 0.973×100.

a State in which direction the decimal point must be moved to calculate the answer.

b State the number of places the decimal point must be moved. Explain your answer.

c Hence calculate the product.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 Consider the fraction $\frac{452}{1000}$.

a Write the fraction as a quotient using a '$\div$' sign.

b 452 is equivalent to 452.0. State in which direction the decimal point in 452.0 must be moved to calculate the answer to part (a).

c State the number of places the decimal point must be moved. Explain your answer.

d Hence write the quotient as a decimal number.

17 Evaluate, writing your answer in decimal form:

a 0.7×10	b 0.0098×10	c 0.02×10	d 0.65×10
e 0.093×10	f 0.3784×10	g 5.2×10	h 0.003×10
i 0.35×100	j 9.3×100	k 6.74×100	l 59.43×100
m 0.035×100	n 0.007×100	o 0.4×100	p 0.0046×100
q 0.204×100	r 0.6489×1000	s 0.04×1000	t 5.761×1000
u 0.008×1000	v 0.0043×1000	w 0.408×1000	x 0.934×1000

18 Evaluate, writing your answer in decimal form:

a $0.3 \div 10$	b $4.707 \div 10$	c $34 \div 10$	d $0.57 \div 10$
e $0.832 \div 10$	f $3.9 \div 10$	g $6.75 \div 10$	h $7.651 \div 10$
i $95.71 \div 10$	j $79.66 \div 100$	k $8.4 \div 100$	l $0.2 \div 100$
m $56 \div 100$	n $423 \div 1000$	o $953.5 \div 100$	p $423.3 \div 1000$
q $5.6 \div 100$	r $0.42 \div 100$	s $7583.5 \div 100$	t $2.2 \div 1000$
u $\frac{52}{100}$	v $\frac{104}{100}$	w $\frac{203}{1000}$	x $\frac{2}{100}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647